MITOCHONDRIA · ASTRONAUT HEALTH INTELLIGENCE

# Health Intelligence Report

Preliminary automated assessment — for Ground Medical review. Not a physician's diagnosis.

| ASTRONAUT | MISSION | ASSESSMENT PERIOD | EXAMINATIONS WITH DATA |
|---|---|---|---|
| Alex Morgan — Synthetic Astronaut | MITOCHONDRIA 2026 - Synthetic Demonstration | 2026-11-01 to 2026-11-08 | 2026-11-05, 2026-11-06, 2026-11-07, 2026-11-08 |

**SYNTHETIC DEMONSTRATION DATA — not real astronaut data. Age, sex, symptoms, medications and history are not recorded or assumed.**

**LITERATURE-BASED SYNTHETIC DEMONSTRATION — NOT CLINICALLY VALIDATED**

**Literature criteria and synthetic identity mappings colour results for this demonstration only: not Researcher-approved, never validated alerts, never applied to real data.**

## Most important findings

HIGH PRIORITY MEDICAL REVIEW

Clinical class AMBER (PROVISIONAL)

### Fever and inflammation with abnormal vital signs

Respiratory rate 26 breaths/min; Oxygen saturation 91 %; Systolic blood pressure 88 mmHg; temperature 38.8 °C; hsCRP 28 mg/L; C-reactive protein 18.4 mg/L; IL-6 5.8 → 74 pg/mL; lactate 4.2 mmol/L (7 Nov, 8 Nov).

Why this priority: PP-1: Respiratory rate, Oxygen saturation (SpO2), Systolic blood pressure in a NEWS2 3-point band. PP-2: fever and CRP above 10 mg/L on the same or adjacent dates.

Next: Ground Medical clinical assessment now: symptoms, mentation and a complete repeated set of vital signs.

HIGH PRIORITY MEDICAL REVIEW

Clinical class AMBER (PROVISIONAL) DATA CHECK

### Electrolyte shifts with a prolonged QTc

Sodium 128 → 146 mmol/L; potassium 5.8 → 3.2 mmol/L; QTc 510 ms (7 Nov, 8 Nov).

Why this priority: PP-3: sodium change above 10 mmol/L within one day with provisional abnormal values at both ends; a DATA CHECK also applies. PP-2: QTc above 500 ms with potassium below 3.5 mmol/L within one day (both are recognised torsade de pointes risk factors).

Next: Ground Medical review of electrolytes and ECG: repeat sodium, potassium and magnesium on a new sample and a 12-lead ECG if the clinician judges it appropriate.

HIGH PRIORITY MEDICAL REVIEW

Clinical class NEUTRAL

### Raised D-dimer with raised acute-phase clotting protein

D-dimer 2.8 mg/L FEU; fibrinogen 520 mg/dL, with Oxygen saturation 91 % and Respiratory rate 26 breaths/min (7 Nov).

Why this priority: PP-2: D-dimer above the cut-off converges with SpO2 or respiratory rate in a NEWS2 ≥2-point band within one day.

Next: Ground Medical to assess the clinical probability of venous thromboembolism before any further testing.

DATA CHECK

### 5 further findings

- HbA1c results need verification — data check
- Cardiac function and strain markers — monitoring or further evaluation
- Raised triglycerides with an unfavourable cholesterol profile — monitoring or further evaluation
- Bone-turnover change and low vitamin D — monitoring or further evaluation + data check
- Poor sleep efficiency and reduced heart-rate variability — monitoring or further evaluation

Next: See the Preliminary Medical Assessment.

| | | | | | |
|---|---|---|---|---|---|
| RED | High priority medical review; not a diagnosis | AMBER | Monitoring or further evaluation | DATA CHECK | Purple — verify the measurement first |
| GREEN | Within applicable reference criteria | NEUTRAL | Insufficient classification evidence | (PROVISIONAL) | Literature-based class; not clinically validated |

Clinical classes: 0 Researcher-approved criteria for this mission; PROVISIONAL = documented literature criterion (synthetic demonstration only; never a validated alert); 2 synthetic identity mapping(s) applied (owner-directed). Panel colour = warning priority, separate from each result's clinical class. General-population criteria are not NASA flight limits.

## Priority next steps

1. Ground Medical clinical assessment now: symptoms, mentation and a complete repeated set of vital signs.
2. Ground Medical review of electrolytes and ECG: repeat sodium, potassium and magnesium on a new sample and a 12-lead ECG if the clinician judges it appropriate.
3. Ground Medical to assess the clinical probability of venous thromboembolism before any further testing.

# Executive health overview

4 examination date(s) with 97 observation(s) in 86 biomarker series. 6 body system(s) show flagged or classified results; 6 series require high-priority review and 1 require measurement verification first.

### 1 · What is happening?

8 connected finding pattern(s) for Ground Medical review: fever and inflammation with abnormal vital signs (high priority medical review); electrolyte shifts with a prolonged qtc (high priority medical review); raised d-dimer with raised acute-phase clotting protein (high priority medical review); hba1c results need verification (data check); cardiac function and strain markers (monitoring or further evaluation); raised triglycerides with an unfavourable cholesterol profile (monitoring …

### 2 · Which results are abnormal or concerning?

Classified: Potassium 5.8 / 3.2 mmol/L (AMBER, provisional); Sodium 128 / 146 mmol/L (AMBER, provisional); HDL cholesterol 28 mg/dL (AMBER, provisional); LDL cholesterol 188 mg/dL (AMBER, provisional); Triglycerides 165 / 330 / 184 mg/dL (AMBER, provisional); hsCRP (inflammation) 28 mg/L (AMBER, provisional); Diastolic blood pressure 54 mmHg (AMBER, provisional); Systolic blood pressure 88 mmHg (AMBER, provisional); Vitamin …

### 3 · What increased or decreased?

Increased: Sodium 128 → 146 mmol/L (+14.1 %); Triglycerides 165 → 330 → 184 mg/dL (+11.5 %) (intermediate peak 330); IL-6 (inflammatory signal) 5.8 → 74 pg/mL (+1176 %); Hematocrit 31 → 34 % (+9.7 %); Hemoglobin 10.2 → 11.3 g/dL (+10.8 %); Osteocalcin (bone formation) 3.5 → 31 ng/mL (+786 %). Decreased: Potassium 5.8 → 3.2 mmol/L (-44.8 %); HbA1c (3-month average blood sugar) 9.1 → 5.9 % (-35.2 %); Heart-rate variability (RMSSD) 24 → 12 ms (-50.0 %).

### 4 · Which changes are biologically meaningful?

Plausible and important if confirmed: Potassium, Sodium, Triglycerides, IL-6 (inflammatory signal), Osteocalcin (bone formation), Heart-rate variability (RMSSD). Verify the measurement first: HbA1c (3-month average blood sugar).

### 5 · Which body systems are affected?

Fluids & electrolytes; Glucose & lipids; Inflammation & immunity; Heart & circulation; Bone & mineral; Sleep & recovery

### 6 · How are the biomarkers related?

Fever and inflammation with abnormal vital signs – Raised triglycerides with an unfavourable cholesterol profile: Inflammation raises plasma triglycerides (established; Khovidhunkit 2004); Fever and inflammation with abnormal vital signs – Raised D-dimer with raised acute-phase clotting protein: Acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing (established; Gabay 1999); Fever and …

### 7 · Which conditions could explain the findings?

acute infection with a systemic inflammatory response (no source is recorded).; non-infectious inflammation, such as tissue injury or strenuous exertion.; a measurement or transcription error affecting several vital signs (less likely if they come from independent devices; devices are not recorded).; pre-analytical or sample-handling error (haemolysis, line contamination, sample mix-up).; a true shift from fluid balance, treatment or illness (treatment and fluid balance are …

### 8 · What supports or contradicts them?

Fever and inflammation with abnormal vital signs: 12 supporting, 1 contradicting/limiting; Electrolyte shifts with a prolonged QTc: 4 supporting, 2 contradicting/limiting; Raised D-dimer with raised acute-phase clotting protein: 4 supporting, 1 contradicting/limiting; HbA1c results need verification: 1 supporting, 0 contradicting/limiting; Cardiac function and strain markers: 4 supporting, 1 contradicting/limiting; Raised triglycerides with an …

### 9 · What should the astronaut do?

Report any symptoms to Ground Medical now (for example fever, breathlessness, chest pain, palpitations or dizziness). Do not change fluid, salt or potassium intake before Ground Medical review. Do not start or change supplements or exercise until Ground Medical has reviewed these findings.

### 10 · What should Ground Medical confirm?

Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time. Do not change fluid, salt or potassium intake before Ground Medical review. Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated.

## Body systems at a glance

### Fluids & electrolytes

AMBER (PROVISIONAL) REVIEW

Potassium 5.8 → 3.2 mmol/L; Sodium 128 → 146 mmol/L; ANP (atrial natriuretic peptide) 35 pg/mL; Blood volume 4.2 L

### Glucose & lipids

AMBER (PROVISIONAL) REVIEW DATA CHECK

HDL cholesterol 28 mg/dL; LDL cholesterol 188 mg/dL; Triglycerides 165 → 330 → 184 mg/dL; ApoB 160 mg/dL

### Inflammation & immunity

AMBER (PROVISIONAL) REVIEW

hsCRP (inflammation) 28 mg/L; IL-6 (inflammatory signal) 5.8 → 74 pg/mL; C-reactive protein 18.4 mg/L; …

### Blood & clotting

aPTT (clotting time) 42 s; D-dimer 2.8 mg/L FEU; Fibrinogen 520 mg/dL; Hematocrit 31 → 34 %

### Heart & circulation

AMBER (PROVISIONAL)

Diastolic blood pressure 54 mmHg; Systolic blood pressure 88 mmHg; Arterial oxygen (PaO2) 65 mmHg; Baroreflex sensitivity …

### Bone & mineral

AMBER (PROVISIONAL) REVIEW

Osteocalcin (bone formation) 3.5 → 31 ng/mL; Vitamin D (25-hydroxy) 12 → 12 ng/mL; Bone failure load (strength) 2100 …

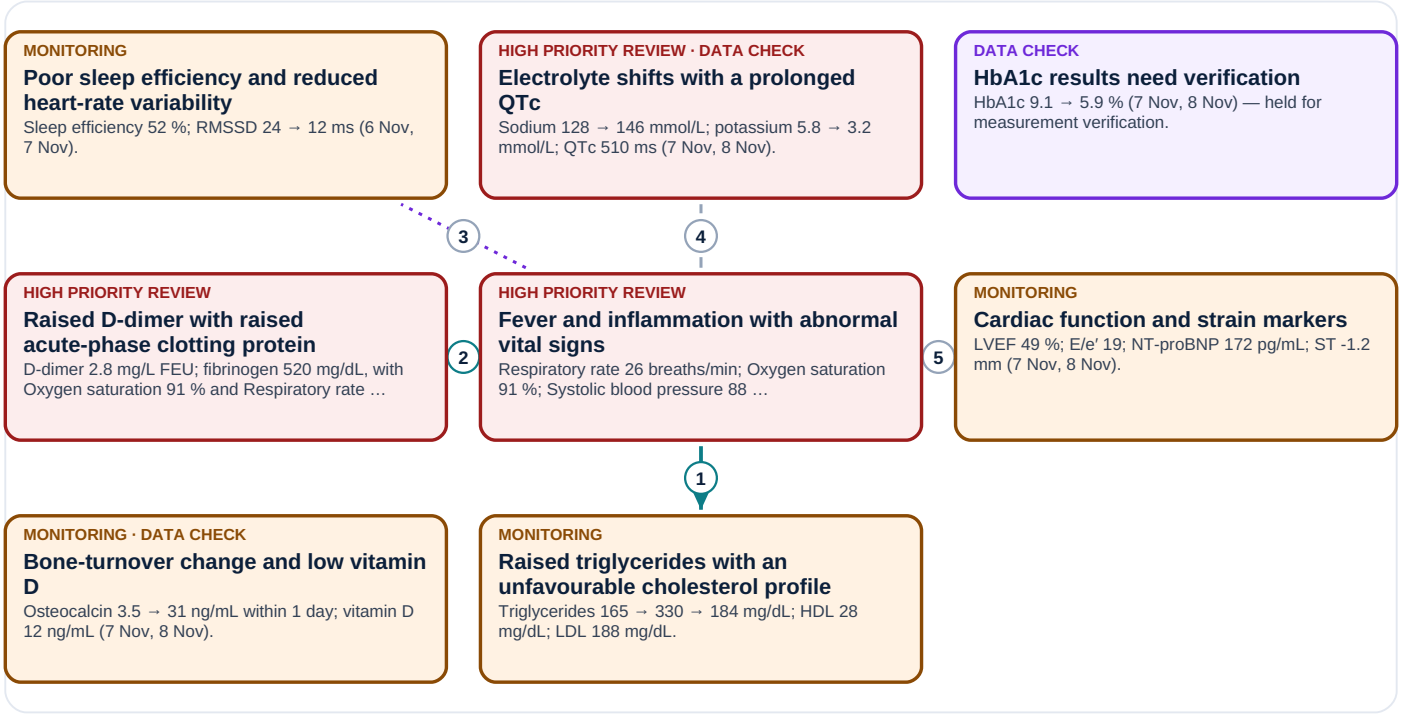
### Sleep & recovery

AMBER (PROVISIONAL) REVIEW

Heart-rate variability (RMSSD) 24 → 12 ms; Sleep efficiency 52 %; Heart-rate variability (SDNN) 18 ms; RR interval 455 ms

## Integrated health assessment

Findings are read as connected patterns. Box colour = warning priority. Solid teal arrows: established biological links with a source. Dashed grey: co-occurrence in this period (not cause). Dotted violet: hypothesis. Numbers refer to the legend below the map.




1. Fever and inflammation with abnormal vital signs — Raised triglycerides with an unfavourable cholesterol profile. Established: Inflammation raises plasma triglycerides (Khovidhunkit 2004).
2. Fever and inflammation with abnormal vital signs — Raised D-dimer with raised acute-phase clotting protein. Established: Acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing (Gabay 1999).
3. Fever and inflammation with abnormal vital signs — Poor sleep efficiency and reduced heart-rate variability. Hypothesis: Illness may reduce sleep quality and HRV.
4. Fever and inflammation with abnormal vital signs — Electrolyte shifts with a prolonged QTc. Co-occurrence: Same dates — co-occurrence, not cause.
5. Fever and inflammation with abnormal vital signs — Cardiac function and strain markers. Co-occurrence: Cardiac findings measured during the acute pattern.

### Overall picture

3 pattern(s) warrant high-priority Ground Medical review; the leading one is fever and inflammation with abnormal vital signs. Part of the picture may follow from it through established biology: raised triglycerides with an unfavourable cholesterol profile (inflammation raises plasma triglycerides; Khovidhunkit 2004); raised D-dimer with raised acute-phase clotting protein (acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing; Gabay 1999). Other findings occur at the same time without an established causal link: poor sleep efficiency and reduced heart-rate variability; electrolyte shifts with a prolonged QTc; cardiac function and strain markers — co-occurrence does not show cause.

Before interpretation, measurements must be verified for: electrolyte shifts with a prolonged QTc; HbA1c results need verification; bone-turnover change and low vitamin D. A data check never lowers a medical-review priority.

All patterns are preliminary, rest on synthetic demonstration data and literature thresholds that are not Researcher-approved, and require clinical evaluation; none is a diagnosis.

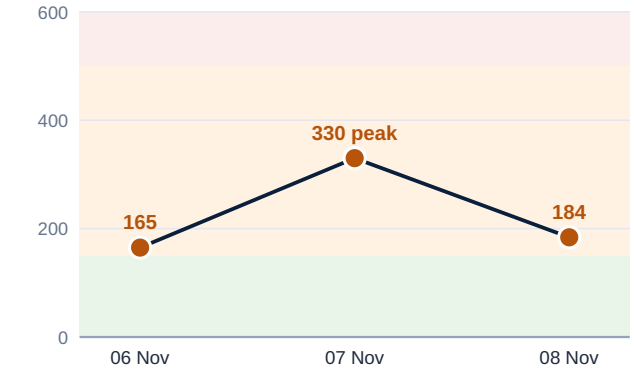
# Longitudinal biomarker intelligence

Every chart shows only observed values, on their actual dates and in their original units. Coloured bands appear only where an approved (or, in demonstration mode, a provisional literature) criterion applies. Hollow violet markers and dashed lines: measurement must be verified first. Grey: unclassified.

## Triglycerides

mg/dL

REVIEW



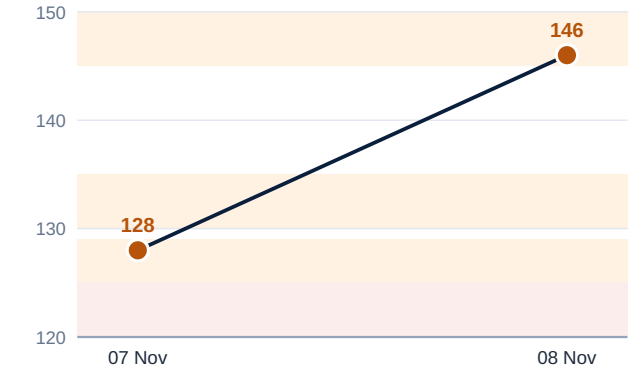

- Healthy (NHLBI, adults): <150 mg/dL
- Borderline high (NHLBI): 150–199 mg/dL
- High (NHLBI): 200–499 mg/dL
- Very high (NHLBI): >500 mg/dL

Triglycerides. 165 → 330 → 184 mg/dL from 2026-11-06 to 2026-11-08 (3 examinations, 2 days). Peaked at 330 mg/dL on 2026-11-07 (intermediate maximum), then fell again; first to last +19 mg/dL (+11.5%). No approved reference criterion applies, so no reference band is drawn. Bands: provisional literature criteria (synthetic demonstration).

## Sodium

mmol/L

REVIEW



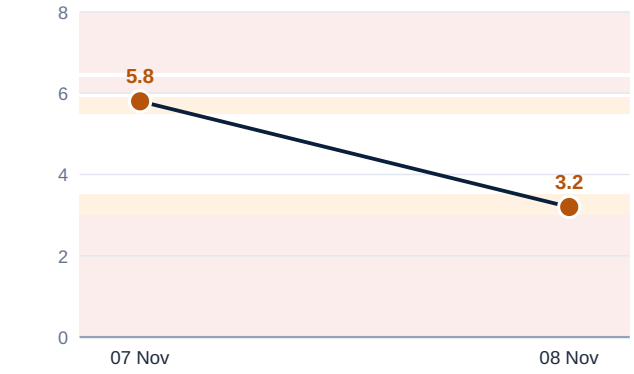

- Mild hyponatraemia (biochemical, ESE): ≥130 and <135 mmol/L
- Moderate hyponatraemia (biochemical, ESE): 125–129 mmol/L
- Profound hyponatraemia (biochemical, ESE): <125 mmol/L
- Hypernatraemia (StatPearls): >145 mmol/L

Sodium. Increased from 128 to 146 mmol/L (+14.1%) between 2026-11-07 and 2026-11-08 (1 day). No approved reference criterion applies, so no reference band is drawn. Bands: provisional literature criteria (synthetic demonstration).

## Potassium

mmol/L

REVIEW



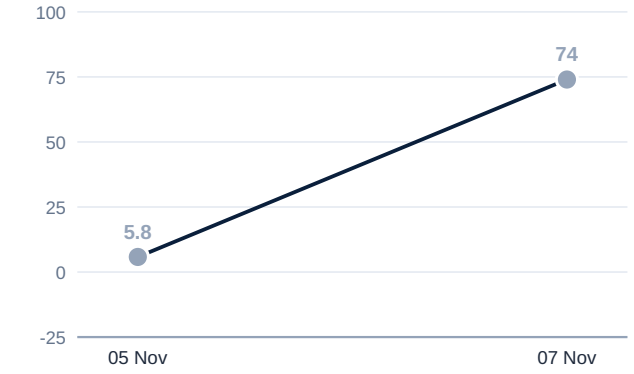

- Mild hyperkalaemia (UKKA): 5.5–5.9 mmol/L
- Moderate hyperkalaemia (UKKA): 6–6.4 mmol/L
- Severe hyperkalaemia (UKKA): ≥6.5 mmol/L
- Mild hypokalaemia (StatPearls): ≥3 and <3.5 mmol/L
- Moderate hypokalaemia (StatPearls): ≥2.5 and <3 mmol/L
- Severe hypokalaemia (StatPearls): <2.5 mmol/L

Potassium. Decreased from 5.8 to 3.2 mmol/L (-44.8%) between 2026-11-07 and 2026-11-08 (1 day). No approved reference criterion applies, so no reference band is drawn. Bands: provisional literature criteria (synthetic demonstration).

## IL-6 (inflammatory signal)

pg/mL

REVIEW

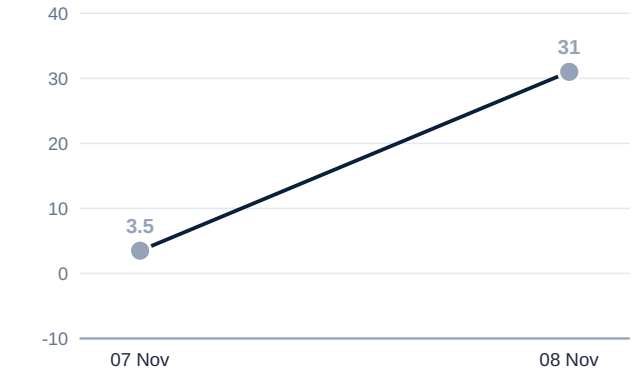



IL-6 (inflammatory signal). Increased from 5.8 to 74 pg/mL (+1175.9%) between 2026-11-05 and 2026-11-07 (2 days). No approved reference criterion applies, so no reference band is drawn.

## Osteocalcin (bone formation)

ng/mL

REVIEW

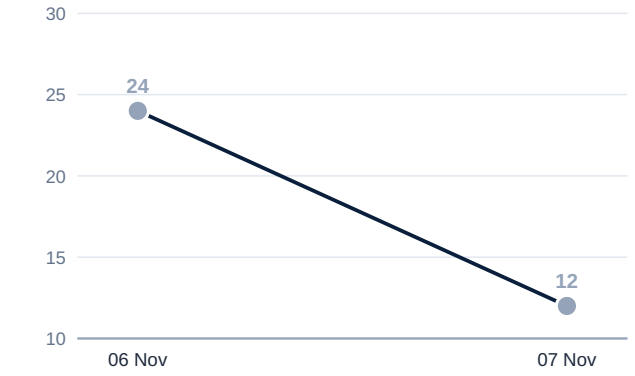



Osteocalcin (bone formation). Increased from 3.5 to 31 ng/mL (+785.7%) between 2026-11-07 and 2026-11-08 (1 day). No approved reference criterion applies, so no reference band is drawn.

## Heart-rate variability (RMSSD)

ms

REVIEW

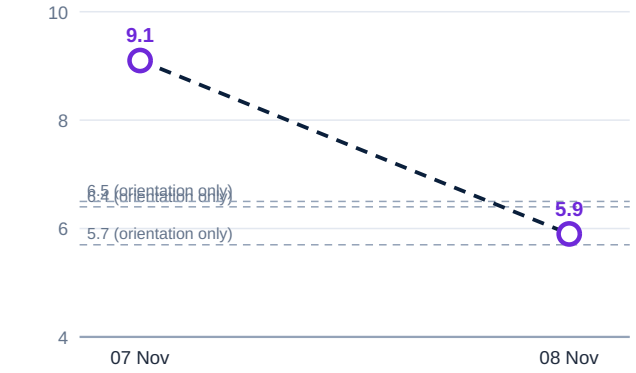



Heart-rate variability (RMSSD). Decreased from 24 to 12 ms (-50%) between 2026-11-06 and 2026-11-07 (1 day). No approved reference criterion applies, so no reference band is drawn.

## HbA1c (3-month average …

%

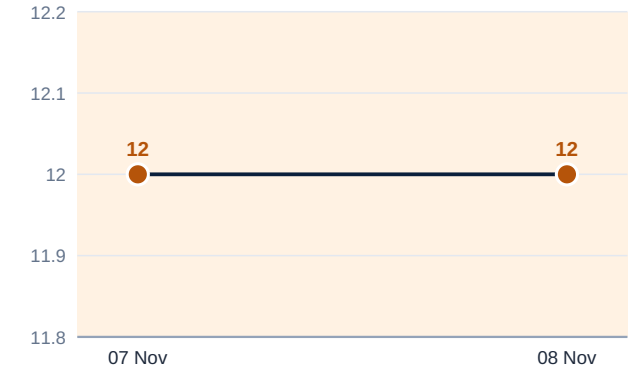
DATA CHECK




HbA1c (3-month average blood sugar). Decreased from 9.1 to 5.9 % (-35.2%) between 2026-11-07 and 2026-11-08 (1 day). 9.1 → 5.9 % (-35.2%) from 2026-11-07 to 2026-11-08: a 1-day interval is far shorter than the period HbA1c reflects; measurement consistency must be checked before interpretation. Bands: provisional literature criteria (synthetic demonstration).

## Vitamin D (25-hydroxy)

ng/mL

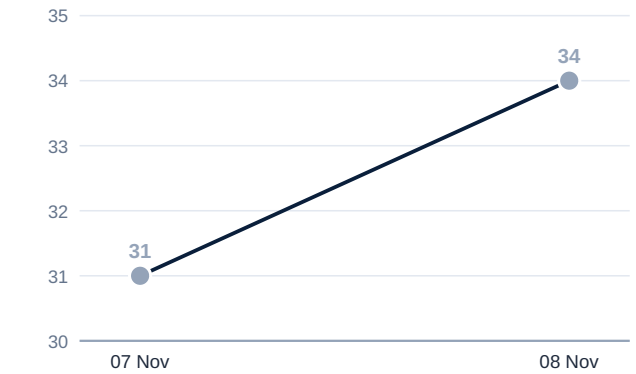



Associated with deficiency (NIH ODS): <12 ng/mL
Generally considered inadequate (NIH ODS): ≥12 and <20 ng/mL
Generally considered adequate (NIH ODS): 20–50 ng/mL
Linked to potential adverse effects (NIH ODS): >50 ng/mL

Vitamin D (25-hydroxy). No recorded change across 2 examinations (12 ng/mL, 2026-11-07 to 2026-11-08). No approved reference criterion applies, so no reference band is drawn. Bands: provisional literature criteria (synthetic demonstration).

## Hematocrit

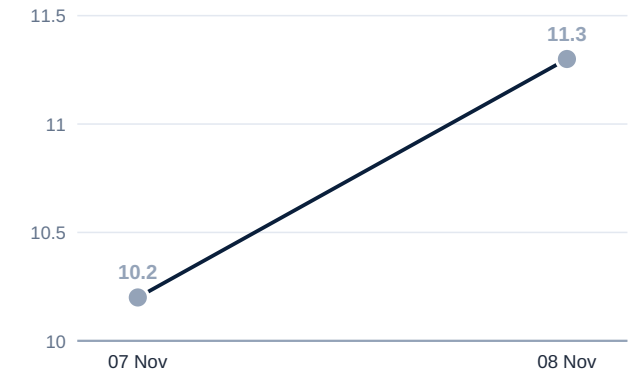
%




Hematocrit. Increased from 31 to 34 % (+9.7%) between 2026-11-07 and 2026-11-08 (1 day). No approved reference criterion applies, so no reference band is drawn.

## Hemoglobin

g/dL




Hemoglobin. Increased from 10.2 to 11.3 g/dL (+10.8%) between 2026-11-07 and 2026-11-08 (1 day). No approved reference criterion applies, so no reference band is drawn.

### Inflammation & immunity — single observations

**hsCRP (inflammation)**
2026-11-07
28 mg/L

### Bone & mineral — single observations

**CTX (bone breakdown)**
2026-11-07
1.82 ng/mL

**P1NP (bone formation)**
2026-11-07
12 ng/mL

3 further unflagged, unclassified result(s) are shown in the results table only.

# Preliminary Medical Assessment — For Ground Medical Review

Automated preliminary assessment generated from the recorded measurements. It lists possible conditions for a qualified clinician to evaluate; it is not a physician-certified diagnosis and assigns no probabilities. Symptoms, examination findings, medications, age and sex were not available unless shown.

## 1 · Fever and inflammation with abnormal vital signs

HIGH PRIORITY MEDICAL REVIEW

| | |
|---|---|
| **WARNING PRIORITY** | HIGH PRIORITY MEDICAL REVIEW (rule PP-1, PP-2). PP-1: Respiratory rate, Oxygen saturation (SpO2), Systolic blood pressure in a NEWS2 3-point band. PP-2: fever and CRP above 10 mg/L on the same or adjacent dates. |
| **CLINICAL CLASS** | • Systolic blood pressure 88 mmHg (11/07): AMBER (PROVISIONAL) — Below 90 mmHg (NHLBI low blood pressure)<br>• hsCRP (inflammation) 28 mg/L (11/07): AMBER (PROVISIONAL) — Above 10 mg/L — repeat and evaluate (CDC/AHA) |
| **OBSERVED** | • Respiratory rate 26 breaths/min on 7 Nov (NEWS2 3-point band; RCP 2017).<br>• Oxygen saturation (SpO2) 91 % on 7 Nov (NEWS2 3-point band; RCP 2017).<br>• Systolic blood pressure 88 mmHg on 7 Nov (NEWS2 3-point band; RCP 2017).<br>• Systolic blood pressure 97 mmHg on 8 Nov (NEWS2 2-point band; RCP 2017).<br>• Body temperature 38.8 °C on 7 Nov — at or above 38 °C, the CDC fever definition.<br>• hsCRP (inflammation) 28 mg/L on 7 Nov — above 10 mg/L; CDC/AHA advise repeating the test and looking for a source of infection or inflammation.<br>• C-reactive protein 18.4 mg/L on 8 Nov — above 10 mg/L; CDC/AHA advise repeating the test and looking for a source of infection or inflammation.<br>• IL-6 5.8 pg/mL → 74 pg/mL (5 Nov → 7 Nov) — review-priority change; no criterion classifies IL-6.<br>• Lactate 4.2 mmol/L on 7 Nov — above 2 mmol/L (Sepsis-3, 2016); interpret with the sampling conditions.<br>• Arterial oxygen (PaO2) 65 mmHg on 7 Nov — recorded; no criterion is applied in this report.<br>• SpO2 91 % on 7 Nov is below the 94–98 % target range used for acutely ill adults (BTS 2017). |
| **MEASUREMENT QUALITY** | • Measurements carry calendar dates only; whether the vital signs were taken at the same time is not established.<br>• Pulse oximetry can read falsely low with poor perfusion, motion or cold extremities; the device and probe site are not recorded.<br>• Temperature measurement site and method are not recorded.<br>• Lactate rises with exertion, tourniquet time and delayed analysis; sampling conditions are not recorded.<br>• CRP results come from different identities (high-sensitivity and standard assays); they are listed separately and not compared as a trend. |
| **CLINICAL PATTERN** | Fever, raised inflammatory markers, vital signs in NEWS2 warning bands and a raised lactate were recorded together on 7 Nov and 8 Nov. Together they describe an acute systemic physiological stress pattern that warrants prompt clinical assessment, including assessment for infection. This is a pattern for review, not a diagnosis. |
| **HYPOTHESES** | • Acute infection with a systemic inflammatory response (no source is recorded).<br>• Non-infectious inflammation, such as tissue injury or strenuous exertion.<br>• A measurement or transcription error affecting several vital signs (less likely if they come from independent devices; devices are not recorded). |
| **ALSO SUPPORTING** | • 7 Nov: respiratory rate 26 breaths/min and systolic pressure 88 mmHg meet 2 of the 3 qSOFA criteria (mentation not recorded) (Sepsis-3, 2016). |
| **CONTRADICTING / LIMITING** | • Systolic blood pressure 97 mmHg on 8 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. |
| **ALTERNATIVES (P58)** | • Intense exercise before sampling.<br>• Sample handling (IL-6 can change with processing delay).<br>• Assay differences between examinations. |
| **MISSING CONTEXT** | • Level of consciousness / mentation and supplemental oxygen — needed for NEWS2 and qSOFA.<br>• Symptoms, examination findings and time course.<br>• White-cell count and differential — not recorded in this period.<br>• qSOFA is not recommended as a single screening tool (SSC 2021); it is shown only as context. |

## 1 · Fever and inflammation with abnormal vital signs (continued)

HIGH PRIORITY MEDICAL REVIEW

| | |
|---|---|
| **CONFIRM** | • Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time.<br>• If Ground Medical judges it appropriate: repeat CRP and a full blood count, and investigate a possible source of infection according to the clinical findings.<br>• Record supplemental oxygen and consciousness so that an aggregate NEWS2 score can be calculated. |
| **SOURCES · GM STATUS** | NEWS2, RCP 2017; CDC; CDC/AHA 2003; Sepsis-3, 2016; SSC 2021; BTS 2017. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 2 · Electrolyte shifts with a prolonged QTc

DATA CHECK HIGH PRIORITY MEDICAL REVIEW

| | |
|---|---|
| **WARNING PRIORITY** | HIGH PRIORITY MEDICAL REVIEW (rule PP-3, PP-2). PP-3: sodium change above 10 mmol/L within one day with provisional abnormal values at both ends; a DATA CHECK also applies. PP-2: QTc above 500 ms with potassium below 3.5 mmol/L within one day (both are recognised torsade de pointes risk factors). A DATA CHECK also applies; it does not lower this priority. |
| **CLINICAL CLASS** | • Sodium 128 mmol/L (11/07): AMBER (PROVISIONAL) — Moderate hyponatraemia (biochemical, ESE)<br>• Sodium 146 mmol/L (11/08): AMBER (PROVISIONAL) — Hypernatraemia (StatPearls)<br>• Potassium 5.8 mmol/L (11/07): AMBER (PROVISIONAL) — Mild hyperkalaemia (UKKA)<br>• Potassium 3.2 mmol/L (11/08): AMBER (PROVISIONAL) — Mild hypokalaemia (StatPearls) |
| **OBSERVED** | • Sodium 128 → 146 mmol/L (7 Nov, 8 Nov): 128 — Moderate hyponatraemia (biochemical, ESE); 146 — Hypernatraemia (StatPearls) (provisional).<br>• Potassium 5.8 → 3.2 mmol/L (7 Nov, 8 Nov): 5.8 — Mild hyperkalaemia (UKKA); 3.2 — Mild hypokalaemia (StatPearls) (provisional).<br>• Sodium changed by 18 mmol/L between 7 Nov and 8 Nov; a rise of more than 10 mmol/L in 24 hours is the limit used to avoid osmotic demyelination when hyponatraemia is corrected (Society for Endocrinology 2022).<br>• QTc 510 ms on 7 Nov — above 500 ms, regarded as highly abnormal and associated with a higher risk of torsade de pointes (AHA/ACCF 2010). |
| **MEASUREMENT QUALITY** | • A high potassium followed by a low value within one day is typical of in-vitro haemolysis or delayed processing of the first sample (pseudohyperkalaemia) (StatPearls); the haemolysis index is not recorded.<br>• A sodium change of this size without documented treatment first requires checking sample identity, intravenous-line contamination and the analyser method (direct or indirect ion-selective electrode).<br>• The QT correction formula and the heart rate at the time of the ECG are not recorded; Bazett's formula overestimates QTc at faster heart rates, so manual re-measurement matters. |
| **CLINICAL PATTERN** | Sodium and potassium moved in opposite directions across the abnormal range within about one day. Low potassium (3.2 mmol/L on 8 Nov) and a QTc of 510 ms are both recognised risk factors for torsade de pointes; they were recorded on different days, so their co-existence is not established. If confirmed, these changes matter for heart rhythm and neurological safety; the measurements alone do not establish a cause. |
| **HYPOTHESES** | • Pre-analytical or sample-handling error (haemolysis, line contamination, sample mix-up).<br>• A true shift from fluid balance, treatment or illness (treatment and fluid balance are not recorded).<br>• QT-prolonging medicines (medication list not recorded). |
| **CONTRADICTING / LIMITING** | • The size and speed of the changes make measurement error at least as likely as a physiological change.<br>• Without kidney-function results the electrolyte changes cannot be placed in context. |
| **ALTERNATIVES (P58)** | • Sample hemolysis or delayed processing, which can falsely raise measured potassium.<br>• Contamination from an intravenous line.<br>• Dilution or concentration of the sample.<br>• Laboratory variation between examinations. |
| **MISSING CONTEXT** | • Magnesium — not recorded in this period.<br>• Creatinine / eGFR (kidney function) — not recorded in this period.<br>• Haemolysis index, sample processing time and analyser method.<br>• Fluid intake/output, intravenous therapy and medications.<br>• ECG rhythm, heart rate at recording and QT correction formula. |
| **CONFIRM** | • Do not change fluid, salt or potassium intake before Ground Medical review.<br>• If Ground Medical judges it appropriate: repeat sodium and potassium on a new, promptly processed sample with the haemolysis index, and add magnesium and kidney function.<br>• If Ground Medical judges it appropriate: repeat a 12-lead ECG with manual QT measurement and the correction formula recorded. |
| **SOURCES · GM STATUS** | Society for Endocrinology 2022; StatPearls; AHA/ACCF 2010. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 3 · Raised D-dimer with raised acute-phase clotting protein

HIGH PRIORITY MEDICAL REVIEW

| | |
|---|---|
| WARNING PRIORITY | HIGH PRIORITY MEDICAL REVIEW (rule PP-2). PP-2: D-dimer above the cut-off converges with SpO2 or respiratory rate in a NEWS2 ≥2-point band within one day. |
| CLINICAL CLASS | • NEUTRAL — no applicable criterion classifies the contributing results; the pattern rests on cited thresholds and review rules, not on a clinical class. |
| OBSERVED | • D-dimer 2.8 mg/L FEU on 7 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded.<br>• Fibrinogen 520 mg/dL on 7 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded).<br>• Oxygen saturation (SpO2) 91 % on 7 Nov (NEWS2 3-point band).<br>• Respiratory rate 26 breaths/min on 7 Nov (NEWS2 3-point band). |
| MEASUREMENT QUALITY | • The D-dimer assay, its own cut-off and the sampling conditions are not recorded. |
| CLINICAL PATTERN | A raised D-dimer together with low oxygen saturation or fast breathing is a combination clinicians weigh when estimating the probability of venous thromboembolism, including pulmonary embolism. D-dimer and fibrinogen also rise with infection and inflammation (fibrinogen is an acute-phase protein; Gabay 1999), so these results cannot indicate thrombosis on their own (ESC 2019). |
| HYPOTHESES | • Acute-phase response to the inflammation recorded in the same period.<br>• Venous thromboembolism — to be assessed clinically; not indicated by the measurements alone. |
| CONTRADICTING / LIMITING | • Concurrent inflammation (hsCRP 28 mg/L; C-reactive protein 18.4 mg/L; Body temperature 38.8 °C) is an alternative reason for raised D-dimer and fibrinogen. |
| MISSING CONTEXT | • Clinical probability assessment with a validated score (symptoms, signs, risk factors).<br>• Age (for the age-adjusted D-dimer cut-off) and the assay's own cut-off.<br>• Platelet count — not recorded in this period. |
| CONFIRM | • Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated. |
| SOURCES · GM STATUS | ESC 2019; Gabay 1999. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 4 · HbA1c results need verification

DATA CHECK

| | |
|---|---|
| WARNING PRIORITY | DATA CHECK (rule PP-5). PP-5: every HbA1c result is held by the existing measurement-consistency rule; no clinical statement is made until it is repeated. |
| CLINICAL CLASS | • NEUTRAL — no applicable criterion classifies the contributing results; the pattern rests on cited thresholds and review rules, not on a clinical class. |
| OBSERVED | • HbA1c 9.1 % on 7 Nov → 5.9 % on 8 Nov. |
| MEASUREMENT QUALITY | • HbA1c reflects average glucose over about 2–3 months (ADA Standards of Care), so a change of 3.2 percentage points within 1 day cannot be physiological. |
| CLINICAL PATTERN | The two HbA1c results cannot both be correct; until one is confirmed, neither supports or excludes a disorder of blood-sugar regulation. |
| HYPOTHESES | • Sample mix-up or transcription error.<br>• Different analytical methods.<br>• Altered red-cell turnover affecting HbA1c (haemoglobin 10.2 g/dL → 11.3 g/dL recorded). |
| ALTERNATIVES (P58) | • Sample mix-up or data-entry error, especially when HbA1c changes markedly within days.<br>• Conditions altering red-cell turnover (anemia, blood loss, transfusion, hemolysis) or hemoglobin variants affecting HbA1c.<br>• A non-fasting or random glucose interpreted as fasting. |
| MISSING CONTEXT | • Fasting plasma glucose — not recorded in this period.<br>• HbA1c method and laboratory. |
| CONFIRM | • Repeat HbA1c on a new sample with the method recorded, together with a fasting plasma glucose, if Ground Medical agrees. |
| SOURCES · GM STATUS | ADA Standards of Care. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 5 · Cardiac function and strain markers

MONITORING OR FURTHER EVALUATION

| | |
|---|---|
| WARNING PRIORITY | MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: borderline findings that do not converge (see the library's cardiac convergence rule). |
| CLINICAL CLASS | • NEUTRAL — no applicable criterion classifies the contributing results; the pattern rests on cited thresholds and review rules, not on a clinical class. |
| OBSERVED | • Ejection fraction 49 % on 7 Nov — in the 41–49 % mildly reduced range (ESC 2021).<br>• E/e′ 19 ratio on 7 Nov — above 14, one ASE/EACVI criterion for raised filling pressure; septal, lateral or average e′ is not recorded (ASE/EACVI 2016).<br>• NT-proBNP 172 pg/mL on 8 Nov — above the 125 pg/mL non-acute exclusion threshold but below the 300 pg/mL acute-setting threshold (ESC 2021).<br>• ST-segment change -1.2 mm on 7 Nov — recorded without lead or context; its significance cannot be assessed from the value alone. |
| MEASUREMENT QUALITY | • Echocardiographic method, image quality and loading conditions are not recorded; measurements during an acute illness may not reflect the baseline. |
| CLINICAL PATTERN | Several measures of heart pumping and filling are borderline-abnormal, recorded during the acute physiological pattern, so they may be transient. They are findings for clinical correlation, not a diagnosis of heart failure. |
| HYPOTHESES | • Transient cardiac strain during acute illness.<br>• Pre-existing cardiac change (no baseline echocardiogram is recorded). |
| CONTRADICTING / LIMITING | • NT-proBNP is below the acute-setting threshold, which argues against acute heart failure. |
| MISSING CONTEXT | • Troponin — not recorded in this period.<br>• A baseline echocardiogram for comparison.<br>• Full echocardiographic diastolic assessment (left-atrial volume, tricuspid regurgitation velocity). |
| CONFIRM | • Ground Medical to decide whether a repeat echocardiogram after recovery from the acute illness, and troponin, are needed. |
| SOURCES · GM STATUS | ESC 2021; ASE/EACVI 2016. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 6 · Raised triglycerides with an unfavourable cholesterol profile

MONITORING OR FURTHER EVALUATION

| | |
|---|---|
| WARNING PRIORITY | MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: provisional AMBER lipid values; fasting status is not recorded, so the triglyceride bands cannot support high priority. |
| CLINICAL CLASS | • Triglycerides 165 mg/dL (11/06): AMBER (PROVISIONAL) — Borderline high (NHLBI)<br>• Triglycerides 330 mg/dL (11/07): AMBER (PROVISIONAL) — High (NHLBI)<br>• Triglycerides 184 mg/dL (11/08): AMBER (PROVISIONAL) — Borderline high (NHLBI)<br>• HDL cholesterol 28 mg/dL (11/07): AMBER (PROVISIONAL) — Low HDL (ATP III)<br>• LDL cholesterol 188 mg/dL (11/07): AMBER (PROVISIONAL) — High (ATP III) |
| OBSERVED | • Triglycerides 165 → 330 → 184 mg/dL (6 Nov, 7 Nov, 8 Nov), peak 330 on 7 Nov; Borderline high (NHLBI) / High (NHLBI) (provisional; fasting not recorded).<br>• HDL cholesterol 28 mg/dL on 7 Nov — Low HDL (ATP III) (provisional).<br>• LDL cholesterol 188 mg/dL on 7 Nov — High (ATP III) (provisional). |
| MEASUREMENT QUALITY | • Fasting status and time since the last meal are not recorded; triglycerides rise for several hours after eating. |
| CLINICAL PATTERN | Triglycerides peaked on 7 Nov and fell the next day. A transient peak of this shape fits an acute effect — a recent meal, or the inflammatory response recorded on the same day, which raises plasma triglycerides (Khovidhunkit 2004) — better than a stable lipid disorder. Low HDL and raised LDL, if confirmed when fasting and well, are long-term cardiovascular risk factors. |
| HYPOTHESES | • Inflammation-related rise in triglycerides (acute-phase lipid changes).<br>• Non-fasting samples. |
| CONTRADICTING / LIMITING | • The fall after 7 Nov argues against a stable, severe hypertriglyceridaemia. |
| ALTERNATIVES (P58) | • Non-fasting sample (the most common cause of a single high value).<br>• Recent alcohol intake.<br>• Medications or other metabolic conditions affecting lipids.<br>• Laboratory or sample-handling variation. |
| MISSING CONTEXT | • Fasting status for each lipid sample.<br>• Total cholesterol and non-HDL cholesterol. |
| CONFIRM | • Repeat a fasting lipid profile once the acute illness has resolved, if Ground Medical agrees. |
| SOURCES · GM STATUS | Khovidhunkit 2004. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 7 · Bone-turnover change and low vitamin D

DATA CHECK | MONITORING OR FURTHER EVALUATION

| | |
|---|---|
| WARNING PRIORITY | MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: a review-priority change and/or a provisional AMBER vitamin D value; no threshold supports high priority. A DATA CHECK also applies; it does not lower this priority. |
| CLINICAL CLASS | • Vitamin D (25-hydroxy) 12 ng/mL (11/07): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS)<br>• Vitamin D (25-hydroxy) 12 ng/mL (11/08): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) |
| OBSERVED | • Osteocalcin 3.5 ng/mL on 7 Nov → 31 ng/mL on 8 Nov (8.9-fold rise).<br>• 25-hydroxyvitamin D 12 ng/mL on 7 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS).<br>• 25-hydroxyvitamin D 12 ng/mL on 8 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS).<br>• CTX (bone breakdown) 1.82 ng/mL on 7 Nov — single measurement; no criterion applied.<br>• P1NP (bone formation) 12 ng/mL on 7 Nov — single measurement; no criterion applied. |
| MEASUREMENT QUALITY | • Osteocalcin is unstable after sampling and varies through the day; the IOF-IFCC recommend P1NP and CTX as the reference bone-turnover markers (IOF-IFCC 2011). A change of this size within 1 day should be verified before biological interpretation. |
| CLINICAL PATTERN | A large, rapid osteocalcin change with single measurements of the reference markers cannot yet be read as a change in bone formation. Vitamin D in the range generally considered inadequate is relevant to bone health in long-duration spaceflight and should be reviewed by Ground Medical. |
| HYPOTHESES | • Pre-analytical variation (sample handling, time of day, assay).<br>• Limited sunlight exposure or intake (intake not recorded). |
| CONTRADICTING / LIMITING | • The reference markers (P1NP, CTX) were measured only once, so they cannot confirm or refute the osteocalcin change. |
| ALTERNATIVES (P58) | • Assay or laboratory differences (osteocalcin is assay-dependent).<br>• Circadian and fasting effects on bone markers.<br>• Recent fracture healing or intense loading. |
| MISSING CONTEXT | • Calcium, phosphate and parathyroid hormone measured together with the bone markers.<br>• Sampling time (fasting morning samples are recommended for CTX). |
| CONFIRM | • Repeat bone-turnover markers (P1NP, CTX, osteocalcin) on fasting morning samples together with calcium, phosphate and PTH, if Ground Medical agrees; Ground Medical to review vitamin D adequacy. |
| SOURCES · GM STATUS | IOF-IFCC 2011; NIH ODS. Ground Medical: Pending Ground Medical review (session review — not persisted). |

## 8 · Poor sleep efficiency and reduced heart-rate variability

MONITORING OR FURTHER EVALUATION

| | |
|---|---|
| WARNING PRIORITY | MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: provisional AMBER sleep efficiency and/or a review-priority HRV change; no threshold supports high priority. |
| CLINICAL CLASS | • Sleep efficiency 52 % (11/07): AMBER (PROVISIONAL) — ≤74 % — does not indicate good sleep quality (NSF) |
| OBSERVED | • Sleep efficiency 52 % on 7 Nov — ≤74 % — does not indicate good sleep quality (NSF) (provisional; NSF 2017).<br>• RMSSD 24 ms on 6 Nov → 12 ms on 7 Nov — review-priority change; no criterion classifies HRV. |
| MEASUREMENT QUALITY | • HRV depends strongly on recording length, posture, breathing and time of day (ESC/NASPE 1996); these conditions are not recorded. |
| CLINICAL PATTERN | Disturbed sleep and lower parasympathetic (vagal) activity coincide with the acute physiological pattern; they indicate reduced recovery but do not identify a cause. |
| HYPOTHESES | • Acute illness, fever and stress reducing sleep quality and HRV.<br>• Workload, schedule or environmental disruption (not recorded). |
| ALTERNATIVES (P58) | • Different measurement posture, time of day or device between examinations.<br>• Recent caffeine, exercise or meals before the recording.<br>• Short recording duration. |
| MISSING CONTEXT | • Sleep log, workload and schedule; HRV recording conditions. |
| CONFIRM | • If Ground Medical agrees, repeat HRV under standardised conditions (same time of day, posture and duration) once the acute illness has resolved. |
| SOURCES · GM STATUS | ESC/NASPE 1996; NSF 2017. Ground Medical: Pending Ground Medical review (session review — not persisted). |

# Personalised recommendations

Each recommendation is linked to the finding that generated it. GM marks actions that require a Ground Medical decision. No medication or supplement doses are given.

### Nutrition

- Triglycerides respond strongly to recent sugars, refined carbohydrates and alcohol; recording intake in the 24 hours before sampling helps interpret the result. Dietary changes should be agreed with Ground Medical.
  → Raised blood triglycerides (possible lipid-metabolism concern): Triglycerides 184 mg/dL on 2026-11-08
- GM Ground Medical should review calcium and vitamin D adequacy; any decision about supplementation belongs to Ground Medical.
  → Change in bone-turnover markers (possible bone-health concern): Osteocalcin (bone formation) 31 ng/mL on 2026-11-08
- Record meal timing and content before blood tests. Diets built on vegetables, legumes, whole grains and adequate protein, with fewer refined sugars and sugary drinks, are generally associated with steadier blood glucose; individual changes should follow Ground Medical advice.
  → Possible disturbance of blood-sugar regulation (dysglycemia): HbA1c (3-month average blood sugar) 5.9 % on 2026-11-08
- GM Do not change fluid or salt intake on the basis of these results until Ground Medical has reviewed them; the appropriate direction depends on the cause.
  → Possible electrolyte disturbance (sodium / potassium): Sodium 146 mmol/L on 2026-11-08; Potassium 3.2 mmol/L on 2026-11-08

### Exercise

- GM Follow Ground Medical's advice on physical activity until the acute findings have been assessed; general exercise suggestions are deferred.
  → Fever and inflammation with abnormal vital signs

## Sleep and recovery

• Protect the scheduled sleep period and record sleep duration around the next recordings.
→ Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 12 ms on 2026-11-07

• Record exercise intensity, sleep and any illness symptoms in the days before the next blood test so the inflammatory markers can be placed in context.
→ Possible inflammatory response: IL-6 (inflammatory signal) 74 pg/mL on 2026-11-07

• Review training load and rest days with the exercise team when heart-rate variability falls.
→ Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 12 ms on 2026-11-07

• Record heart-rate variability under the same posture, time of day and device so that changes can be compared.
→ Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 12 ms on 2026-11-07

• Repeat the changed measurements in red blood cells and blood volume with the same laboratory method at the next scheduled examination, so the change can be confirmed.
→ Red blood cells and blood volume (change below the review rules)

• Repeat the single-examination measurements at the next scheduled examination with the same laboratory, method and units, so that changes over time can be measured.
→ Heart and circulation (single examination); Blood clotting and venous flow (single examination)

## Medical follow-up

GM · Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time.
→ Fever and inflammation with abnormal vital signs

GM · If Ground Medical judges it appropriate: repeat CRP and a full blood count, and investigate a possible source of infection according to the clinical findings.
→ Fever and inflammation with abnormal vital signs

GM · Do not change fluid, salt or potassium intake before Ground Medical review.
→ Electrolyte shifts with a prolonged QTc

GM · If Ground Medical judges it appropriate: repeat sodium and potassium on a new, promptly processed sample with the haemolysis index, and add magnesium and kidney function.
→ Electrolyte shifts with a prolonged QTc

GM · Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated.

→ Raised D-dimer with raised acute-phase clotting protein

GM · Repeat HbA1c on a new sample with the method recorded, together with a fasting plasma glucose, if Ground Medical agrees.
→ HbA1c results need verification

GM · Ground Medical to decide whether a repeat echocardiogram after recovery from the acute illness, and troponin, are needed.
→ Cardiac function and strain markers

GM · Repeat a fasting lipid profile once the acute illness has resolved, if Ground Medical agrees.
→ Raised triglycerides with an unfavourable cholesterol profile

GM · Repeat bone-turnover markers (P1NP, CTX, osteocalcin) on fasting morning samples together with calcium, phosphate and PTH, if Ground Medical agrees; Ground Medical to review vitamin D adequacy.
→ Bone-turnover change and low vitamin D

GM · If Ground Medical agrees, repeat HRV under standardised conditions (same time of day, posture and duration) once the acute illness has resolved.
→ Poor sleep efficiency and reduced heart-rate variability

GM · Check sample collection and handling, fluid intake and urine output, and kidney function (creatinine or eGFR) alongside the repeat measurements.
→ Fluids, electrolytes and kidney function — Sodium and Potassium (review flag)

# Complete results — all observations

97 observations in 86 biomarker series from 4 examination date(s), exactly as recorded (no unit conversion). Classes use approved criteria or, where marked PROVISIONAL, literature criteria for this synthetic demonstration; review priority and data quality are shown separately.

| Measurement | Value | Date | Class | Review / data | Criterion |
|---|---|---|---|---|---|
| **FLUIDS & ELECTROLYTES** | | | | | |
| ANP (atrial natriuretic peptide) | **35 pg/mL** | 11/07 | NEUTRAL | — | — |
| Blood volume | **4.2 L** | 11/07 | NEUTRAL | — | — ◊ |
| Plasma volume | **2.5 L** | 11/07 | NEUTRAL | — | — ◊ |
| Potassium | **5.8 mmol/L** | 11/07 | AMBER PROV. | HIGH | Mild hyperkalaemia (UKKA) (5.5–5.9 mmol/L) ‡ |
| Potassium | **3.2 mmol/L** | 11/08 | AMBER PROV. | HIGH | Mild hypokalaemia (StatPearls) (≥3 and <3.5 mmol/L) |
| Sodium | **128 mmol/L** | 11/07 | AMBER PROV. | HIGH | Moderate hyponatraemia (biochemical, ESE) (125–129 mmol/L) ‡ |
| Sodium | **146 mmol/L** | 11/08 | AMBER PROV. | HIGH | Hypernatraemia (StatPearls) (>145 mmol/L) |
| **GLUCOSE & LIPIDS** | | | | | |
| ApoB | **160 mg/dL** | 11/07 | NEUTRAL | — | — |
| HbA1c (3-month average blood sugar) | **9.1 %** | 11/07 | NEUTRAL | CHECK | Held — verify the measurement first |
| HbA1c (3-month average blood sugar) | **5.9 %** | 11/08 | NEUTRAL | CHECK | Held — verify the measurement first |
| HDL cholesterol | **28 mg/dL** | 11/07 | AMBER PROV. | — | Low HDL (ATP III) (<40 mg/dL) |
| Insulin | **24 µIU/mL** | 11/07 | NEUTRAL | — | — |
| LDL cholesterol | **188 mg/dL** | 11/07 | AMBER PROV. | — | High (ATP III) (160–189 mg/dL) ‡ |
| Triglycerides | **165 mg/dL** | 11/06 | AMBER PROV. | HIGH | Borderline high (NHLBI) (150–199 mg/dL) ‡ |
| Triglycerides | **330 mg/dL** | 11/07 | AMBER PROV. | HIGH | High (NHLBI) (200–499 mg/dL) ‡ |
| Triglycerides | **184 mg/dL** | 11/08 | AMBER PROV. | HIGH | Borderline high (NHLBI) (150–199 mg/dL) ‡ |
| **INFLAMMATION & IMMUNITY** | | | | | |
| C-reactive protein | **18.4 mg/L** | 11/08 | NEUTRAL | — | — ◊ |
| hsCRP (inflammation) | **28 mg/L** | 11/07 | AMBER PROV. | — | Above 10 mg/L — repeat and evaluate (CDC/AHA) (>10 mg/L) |
| IgA (antibodies) | **32 mg/dL** | 11/07 | NEUTRAL | — | — ◊ |
| IgG (antibodies) | **380 mg/dL** | 11/07 | NEUTRAL | — | — ◊ |
| IgM (antibodies) | **25 mg/dL** | 11/07 | NEUTRAL | — | — ◊ |
| IL-10 (anti-inflammatory signal) | **25 pg/mL** | 11/07 | NEUTRAL | — | — ◊ |
| IL-12 | **6 pg/mL** | 11/07 | NEUTRAL | — | — ◊ |

| Measurement | Value | Date | Class | Review / data | Criterion |
|---|---|---|---|---|---|
| IL-2 | **2 pg/mL** | 11/07 | NEUTRAL | — | — ◊ |
| IL-4 | **12 pg/mL** | 11/07 | NEUTRAL | — | — ◊ |
| IL-6 (inflammatory signal) | **5.8 pg/mL** | 11/05 | NEUTRAL | HIGH | — |
| IL-6 (inflammatory signal) | **74 pg/mL** | 11/07 | NEUTRAL | HIGH | — |
| TNF-α (inflammatory signal) | **42 pg/mL** | 11/07 | NEUTRAL | — | — |
| **BLOOD & CLOTTING** | | | | | |
| aPTT (clotting time) | **42 s** | 11/07 | NEUTRAL | — | — |
| D-dimer | **2.8 mg/L FEU** | 11/07 | NEUTRAL | — | — ◊ |
| Fibrinogen | **520 mg/dL** | 11/07 | NEUTRAL | — | — ◊ |
| Hematocrit | **31 %** | 11/07 | NEUTRAL | — | — |
| Hematocrit | **34 %** | 11/08 | NEUTRAL | — | — |
| Hemoglobin | **10.2 g/dL** | 11/07 | NEUTRAL | — | — |
| Hemoglobin | **11.3 g/dL** | 11/08 | NEUTRAL | — | — |
| **HEART & CIRCULATION** | | | | | |
| Arterial oxygen (PaO2) | **65 mmHg** | 11/07 | NEUTRAL | — | — |
| Baroreflex sensitivity | **3 ms/mmHg** | 11/07 | NEUTRAL | — | — |
| Cardiac index | **2.9 L/min/m²** | 11/07 | NEUTRAL | — | — |
| Cardiac output | **5.54 L/min** | 11/07 | NEUTRAL | — | — |
| Carotid wall thickness (cIMT) | **0.95 mm** | 11/07 | NEUTRAL | — | — |
| Central venous pressure | **3 mmHg** | 11/07 | NEUTRAL | — | — |
| Cerebral blood-flow velocity | **35 cm/s** | 11/07 | NEUTRAL | — | — |
| Diastolic blood pressure | **54 mmHg** | 11/07 | AMBER PROV. | — | Below 60 mmHg (NHLBI low blood pressure) (<60 mmHg) |
| Diastolic blood pressure | **61 mmHg** | 11/08 | NEUTRAL | — | — ◊ |
| Ejection fraction (LVEF) | **49 %** | 11/07 | NEUTRAL | — | — |
| End-tidal CO2 | **28 mmHg** | 11/07 | NEUTRAL | — | — |
| Exercise workload | **45 W** | 11/07 | NEUTRAL | — | — |
| Filling pressure ratio E/e′ | **19 ratio** | 11/07 | NEUTRAL | — | — |
| Filling ratio E/A | **0.6 ratio** | 11/07 | NEUTRAL | — | — |
| Flow-mediated dilation (vessel function) | **2.5 %** | 11/07 | NEUTRAL | — | — |
| Jugular vein peak velocity | **9 cm/s** | 11/07 | NEUTRAL | — | — |
| Lactate | **4.2 mmol/L** | 11/07 | NEUTRAL | — | — |
| Left-ventricle end-diastolic volume | **86 mL** | 11/07 | NEUTRAL | — | — |
| Left-ventricle end-systolic volume | **44 mL** | 11/07 | NEUTRAL | — | — |
| Left-ventricle mass | **125 g** | 11/07 | NEUTRAL | — | — |
| Mean arterial pressure | **65.3 mmHg** | 11/07 | NEUTRAL | — | — |
| NT-proBNP (heart strain) | **172 pg/mL** | 11/08 | NEUTRAL | — | — ◊ |
| Orthostatic change in systolic pressure | **-28 mmHg** | 11/07 | NEUTRAL | — | — |
| Orthostatic heart rate change — SYNTHETIC DEMO (not scientifically validated) | **34 bpm** | 11/08 | NEUTRAL | — | — ◊ |
| Oxygen saturation (SpO2) | **91 %** | 11/07 | NEUTRAL | — | — |
| PR interval | **120 ms** | 11/07 | NEUTRAL | — | — |
| Pulse pressure | **34 mmHg** | 11/07 | NEUTRAL | — | — |
| Pulse-wave velocity (arterial stiffness) | **12.4 m/s** | 11/07 | NEUTRAL | — | — |

| Measurement | Value | Date | Class | Review / data | Criterion |
|---|---|---|---|---|---|
| QRS duration | **88 ms** | 11/07 | NEUTRAL | — | — |
| QTc interval | **510 ms** | 11/07 | NEUTRAL | — | — |
| Resting heart rate | **94 bpm** | 11/08 | NEUTRAL | — | — ◊ |
| ST-segment change | **-1.2 mm** | 11/07 | NEUTRAL | — | — |
| Stroke volume | **42 mL/beat** | 11/07 | NEUTRAL | — | — |
| Systemic vascular resistance | **1100 dyn·s/cm^5** | 11/07 | NEUTRAL | — | — |
| Systolic blood pressure | **88 mmHg** | 11/07 | AMBER PROV. | — | Below 90 mmHg (NHLBI low blood pressure) (<90 mmHg) |
| Systolic blood pressure | **97 mmHg** | 11/08 | NEUTRAL | — | — ◊ |
| Venous return | **4.8 L/min** | 11/07 | NEUTRAL | — | — |
| VO2peak (exercise capacity) | **19 mL/kg/min** | 11/07 | NEUTRAL | — | — |
| **BONE & MINERAL** | | | | | |
| Bone failure load (strength) | **2100 N** | 11/07 | NEUTRAL | — | — ◊ |
| CTX (bone breakdown) | **1.82 ng/mL** | 11/07 | NEUTRAL | — | — ◊ |
| Deoxypyridinoline (bone breakdown) | **38 nmol/mmol creatinine** | 11/07 | NEUTRAL | — | — ◊ |
| Dietary calcium intake | **420 mg/day** | 11/07 | NEUTRAL | — | — ◊ |
| Femoral neck bone density | **0.54 g/cm²** | 11/07 | NEUTRAL | — | — ◊ |
| Osteocalcin (bone formation) | **3.5 ng/mL** | 11/07 | NEUTRAL | HIGH | — ◊ |
| Osteocalcin (bone formation) | **31 ng/mL** | 11/08 | NEUTRAL | HIGH | — ◊ |
| P1NP (bone formation) | **12 ng/mL** | 11/07 | NEUTRAL | — | — ◊ |
| Pyridinoline (bone breakdown) | **165 nmol/mmol creatinine** | 11/07 | NEUTRAL | — | — ◊ |
| Total hip bone density | **0.61 g/cm²** | 11/07 | NEUTRAL | — | — ◊ |
| Vitamin D (25-hydroxy) | **12 ng/mL** | 11/07 | AMBER PROV. | — | Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊ |
| Vitamin D (25-hydroxy) | **12 ng/mL** | 11/08 | AMBER PROV. | — | Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊ |
| **SLEEP & RECOVERY** | | | | | |
| Heart-rate variability (RMSSD) | **24 ms** | 11/06 | NEUTRAL | HIGH | — |
| Heart-rate variability (RMSSD) | **12 ms** | 11/07 | NEUTRAL | HIGH | — |
| Heart-rate variability (SDNN) | **18 ms** | 11/07 | NEUTRAL | — | — |
| RR interval | **455 ms** | 11/07 | NEUTRAL | — | — |
| Sleep efficiency | **52 %** | 11/07 | AMBER PROV. | — | ≤74 % — does not indicate good sleep quality (NSF) (≤74 %) § ◊ |
| **OTHER MEASUREMENTS** | | | | | |
| Body temperature | **38.8 °C** | 11/07 | NEUTRAL | — | — ◊ |
| G-CSF — SYNTHETIC DEMO (not scientifically validated) | **12.5 pg/mL** | 11/07 | NEUTRAL | — | — ◊ |
| Mission Duration — SYNTHETIC DEMO (not scientifically validated) | **118 days** | 11/07 | NEUTRAL | — | — ◊ |
| Respiratory rate | **26 breaths/min** | 11/07 | NEUTRAL | — | — ◊ |
| Trabecular number | **0.7 1/mm** | 11/07 | NEUTRAL | — | — ◊ |
| Trabecular separation | **2.1 mm** | 11/07 | NEUTRAL | — | — ◊ |
| Urine pH — SYNTHETIC DEMO (not scientifically validated) | **5.1 pH** | 11/07 | NEUTRAL | — | — ◊ |

— No applicable criterion: NEUTRAL (insufficient classification evidence).

‡ Per-measurement conditions of the provisional criterion that are not recorded: Haemolysis, prolonged transit and delayed separation are recorded as excluded for this sample; The specimen type (serum or lithium-heparin plasma) is recorded for this sample; The observation records that sodium was measured by ion-specific electrode; Hyperglycaemia is recorded as excluded for this sample (or a glucose-corrected sodium is recorded); The observation records the fasting state of the sample; The

observation records the fasting state (fasting time) of the sample; The observation records the 25(OH)D assay method.

§ Identity bound through an owner-directed synthetic demonstration mapping (not Researcher-approved).

◊ Synthetic demonstration catalog identity (not scientifically validated).

# Methodology note

• Data source and census. Read from this device's encrypted local store: 235 record(s) read of 235 stored (complete). Census p60-observation-census.v1: 97 observations, 86 series, dates 2026-11-05, 2026-11-06, 2026-11-07, 2026-11-08; SHA-256 bbb957144670d5881e9aca9eabc97d19f55a0fed4419dd6fcf6fde074139792f. Records held only on another device or only in the cloud are not included unless imported through the explicit cross-device sync. Generator p60-premium-report.v1 · Evidence ID MITO-EV-7D83-9BC1-F8D0-8EDD.

• Data. 97 observations · 86 biomarker series (exact catalog identity + unit) · 4 examination date(s) · 10 series compared between examinations · excluded: 1 Duplicate projection of cf1b5546-d445-4d16-8050-9c5e5ab20045; counted once. Values exactly as recorded; no unit conversion or interpolation. Synthetic demonstration data.

• Calculations. Changes only between consecutive observations of the same identity and unit; the full trajectory, including intermediate peaks and troughs, is reported.

• Criteria. 0 Researcher-approved criteria available. LITERATURE-BASED SYNTHETIC DEMONSTRATION — NOT CLINICALLY VALIDATED: provisional literature interpretation ON (synthetic demonstration only; marked PROVISIONAL; never approved classifications or validated alerts). General-population criteria are not NASA flight limits.

• Synthetic identity mappings. 25 owner-directed mappings re-verified against the repository (same concept, identical unit, same domain, unique target, named mission); not Researcher-approved and never used for approved classification, real data or unit conversion. Used in this report: Vitamin D (25-hydroxy) (demo:ubt-7b25ce69 → criterion:p59-nihods-25ohd, ng/mL); Sleep efficiency (demo:ubt-efe7e2fb → criterion:p59-nsf2017-sleep-efficiency, %).

• Warning priority (panel colour) is a separate axis from clinical class. Every P58 concern in this report is represented by a P60 pattern, so the P59 rules WP-1 to WP-5 are not shown here; they remain in force for concerns that no pattern covers.

• Cross-domain pattern analysis (P60, owner directed synthetic demonstration pending researcher review): 8 pattern(s) from 35 concepts bound by exact identity and unit; priorities by rules PP-1, PP-2, PP-3, PP-4, PP-5. Patterns never change a result's clinical class. Priority rules: PP-1 any vital sign in a NEWS2 3-point band; PP-2 two independent findings converging on the same or adjacent dates; PP-3 sodium change above 10 mmol/L within one day with abnormal values at both ends (plus DATA CHECK); PP-4 single threshold or review change (monitoring); PP-5 all contributing results held (data check).

• Automated assessment. AI-assisted clinical reasoning is switched off for this deployment. The deterministic clinical reasoning engine produced the assessment. Review priority and data quality are separate from clinical classification. Not a diagnosis; no medication or dose advice.

## References for the cross-domain patterns

NEWS2, RCP 2017 — Royal College of Physicians. National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS. Updated report of a working party. London: RCP, 2017.

CDC — Centers for Disease Control and Prevention. Definitions of Signs, Symptoms, and Conditions of Ill Travelers (Port Health).

CDC/AHA 2003 — Pearson TA, Mensah GA, Alexander RW, et al. Markers of inflammation and cardiovascular disease: application to clinical and public health practice. A statement for healthcare professionals from the CDC and the AHA. Circulation. 2003;107(3):499-511.

Sepsis-3, 2016 — Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA. 2016;315(8):801-810.

SSC 2021 — Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Crit Care Med. 2021;49(11):e1063-e1143.

BTS 2017 — O'Driscoll BR, Howard LS, Earis J, Mak V. BTS guideline for oxygen use in adults in healthcare and emergency settings. Thorax. 2017;72(Suppl 1):ii1-ii90.

Society for Endocrinology 2022 — Society for Endocrinology Endocrine Emergency Guidance: Emergency management of severe and moderately severely symptomatic hyponatraemia in adult patients (2022).

StatPearls — Simon LV, Hashmi MF, Farrell MW. Hyperkalemia. StatPearls. Treasure Island (FL): StatPearls Publishing.

AHA/ACCF 2010 — Drew BJ, Ackerman MJ, Funk M, et al. Prevention of torsade de pointes in hospital settings: a scientific statement from the American Heart Association and the American College of Cardiology Foundation. Circulation. 2010;121:1047-1060.

ESC 2019 — Konstantinides SV, Meyer G, Becattini C, et al. 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society. Eur Heart J. 2020;41(4):543-603.

Gabay 1999 — Gabay C, Kushner I. Acute-phase proteins and other systemic responses to inflammation. N Engl J Med. 1999;340(6):448-454.

ADA Standards of Care — American Diabetes Association Professional Practice Committee. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes (section 6).

ESC 2021 — McDonagh TA, Metra M, Adamo M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. Eur Heart J. 2021;42(36):3599-3726.

ASE/EACVI 2016 — Nagueh SF, Smiseth OA, Appleton CP, et al. Recommendations for the evaluation of left ventricular diastolic function by echocardiography: an update from the ASE and EACVI. J Am Soc Echocardiogr. 2016;29(4):277-314.

Khovidhunkit 2004 — Khovidhunkit W, Kim MS, Memon RA, et al. Effects of infection and inflammation on lipid and lipoprotein metabolism: mechanisms and consequences to the host. J Lipid Res. 2004;45(7):1169-1196.

IOF-IFCC 2011 — Vasikaran S, Eastell R, Bruyère O, et al. Markers of bone turnover for the prediction of fracture risk and monitoring of osteoporosis treatment: a need for international reference standards. Osteoporos Int. 2011;22(2):391-420.

NIH ODS — NIH Office of Dietary Supplements. Vitamin D — Fact Sheet for Health Professionals.

ESC/NASPE 1996 — Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability: standards of measurement, physiological interpretation and clinical use. Circulation. 1996;93(5):1043-1065.

NSF 2017 — Ohayon M, Wickwire EM, Hirshkowitz M, et al. National Sleep Foundation's sleep quality recommendations: first report. Sleep Health. 2017;3(1):6-19.